

Causal Inference for Asset Pricing

Haddad, He, Huebner, Kondor, Loualiche

Discussant: Philippe van der Beck¹

¹Harvard Business School

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 - ▶ Holdings constant (other) investors' demand curve, what is the impact of removing 1% of outstanding supply of an asset/portfolio?

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- ▶ Two ways to answer this question:
 1. Write down model of investor demand (finance), consumer demand (economics).
E.g. CARA normal model implies impact given by risk and risk aversion.
 2. Estimate impact directly in reduced form regression.²

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This Paper: An explicit guide for Approach 2.

1. Define the main challenge: Cannot view assets in isolation. $N \times N$ matrix of spillover effects. That's a lot of parameters ☹️. Particularly, for simple cross-sectional regression with N obs.
2. Provide solution:
 - ▶ Can identify *something* (the relative price impact) from single cross-sectional regression
 - ▶ Formal assumption: **Homogenous substitution conditional on observables**

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 - ▶ Different l.h. side, different controls, different instruments, time series, cross-section, panel.
 - ▶ Paper provides very useful **guidance** for this literature:³
What exactly does $\hat{\beta}$ capture and under which assumptions.
 - ▶ Importantly: This paper is not about instruments.

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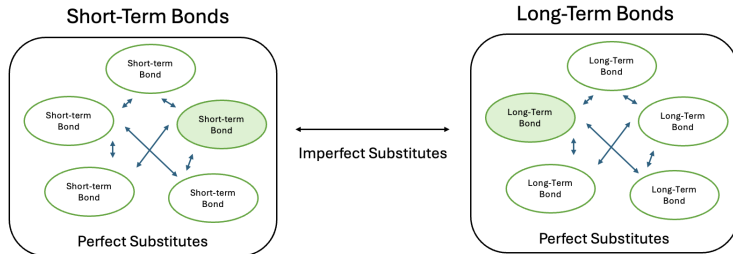
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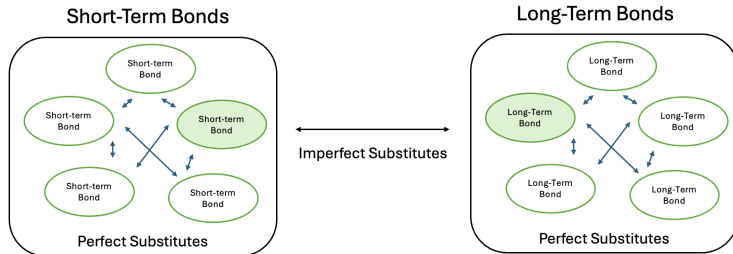
1. Advertise/repeat the main message with simple example.
"Homogenous substitution conditional on observables (HSCO)." What does it mean?
2. Relation to existing literature: What is new and what is not?
3. When do we care about relative multipliers?

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Homogenous Substitution Conditional on Observables

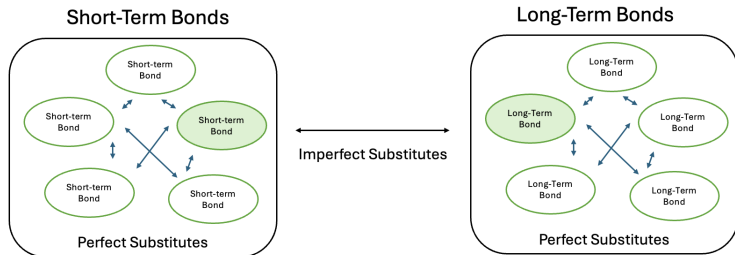


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→ All LT Bonds increase by same amount, all ST Bonds decrease by same amount.
- ▶ Cross-Sectional regression across *all* bonds $n = 1, \dots, N$:

$$\Delta P_n = \alpha + \beta \Delta Q_n^{Fed} + Controls + \epsilon_n$$

1. Without controls: $\hat{\beta} > 0$
Homogenous substitution conditional on observables **violated**
2. Controlling for $Maturity_n$: $\hat{\beta} = 0$.
Homogenous substitution conditional on observables **satisfied**

Motivating Example

The example highlights two key take-aways of the paper.

1. Controlling for stuff changes interpretation of regression coefficients.
 - ▶ we are identifying *relative* effects, relative to what we control for
2. Only if we control for right stuff, the coefficient actually has a structural interpretation.
 - ▶ What's the right stuff? It should capture heterogenous substitution, such that only homogenous substitution is left.
 - ▶ Then the β measures relative impact (relative to assets with similar controls).

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What if we cannot credibly verify “homog. substitution cond. on observables”?

- ▶ Do we have to go back to Shleifer 1986 and say: You haven't controlled for the right stuff?
- ▶ No. Controlling for different characteristics (FF3, industry, beta etc.) doesn't change equity multiplier much. See Chaudhary et al. (2025). Perhaps also show this?
- ▶ This is a pro demand-system paper.

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5. The problem of substitute spillovers in cross-sectional regression of prices and quantities is not specific to finance. Long-standing problem in IO.
 - ▶ You make specific assumption that allows identifying *relative* elasticities.
 - ▶ IO literature makes other assumptions (Deaton & Muellbauer, Berry et al. 1995, etc)
 - ▶ Monardo (2021) substitution across continuous characteristics (Flexible Inverse Logit).

The zoo of terminology and approaches highlights the need for and importance of this paper!!!

Proposal: What about this structure of the paper

1. Title: *Elasticity Estimation (in Finance): What, where, when and how?*
2. What has been done, what assumptions have been made.
3. Why is factor structure in elasticity/multiplier matrix⁴ the best approach (in a finance context)?
4. Theory: Written in terms of multipliers as opposed to elasticities (in line with empirics)
5. Empirical Applications: Bonds *and* equities.

⁴aka Homogenous substitution conditional on observables

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... But did (reduced form) *causal inference* care about identifying the full structural matrix?

Conclusion

- ▶ Very nice paper! Clear guidance for exciting and important literature.
 - ▶ The use of quantities in finance has been killed in the 80s once. Let's not do it again.
 - ▶ "*Substitutes matter in finance*": This paper formalizes the statistical assumption required for estimation.
- ▶ **pro-"using quantities in asset pricing"-paper**

Appendix

Background: Demand-System Asset Pricing

- ▶ Given endowments, preferences, cash flow distribution, any asset pricing model implies
 1. an SDF that prices all assets: $P = E[M \times \text{Cashflows}]$
 2. a demand system $Q_i(P)$ such that $\sum_i Q_i(P) = \text{Supply}$
- ▶ The demand-based approach is useful if portfolio holdings Q_i contain incremental information *beyond* prices P .
- ▶ When is that the case?
 1. Investors are heterogenous: Fact! ✓
 2. Demand elasticity $< \infty$: Strong evidence (✓)
- ▶ Conditional on 1. and 2. being true: How do we exactly link quantities and prices?